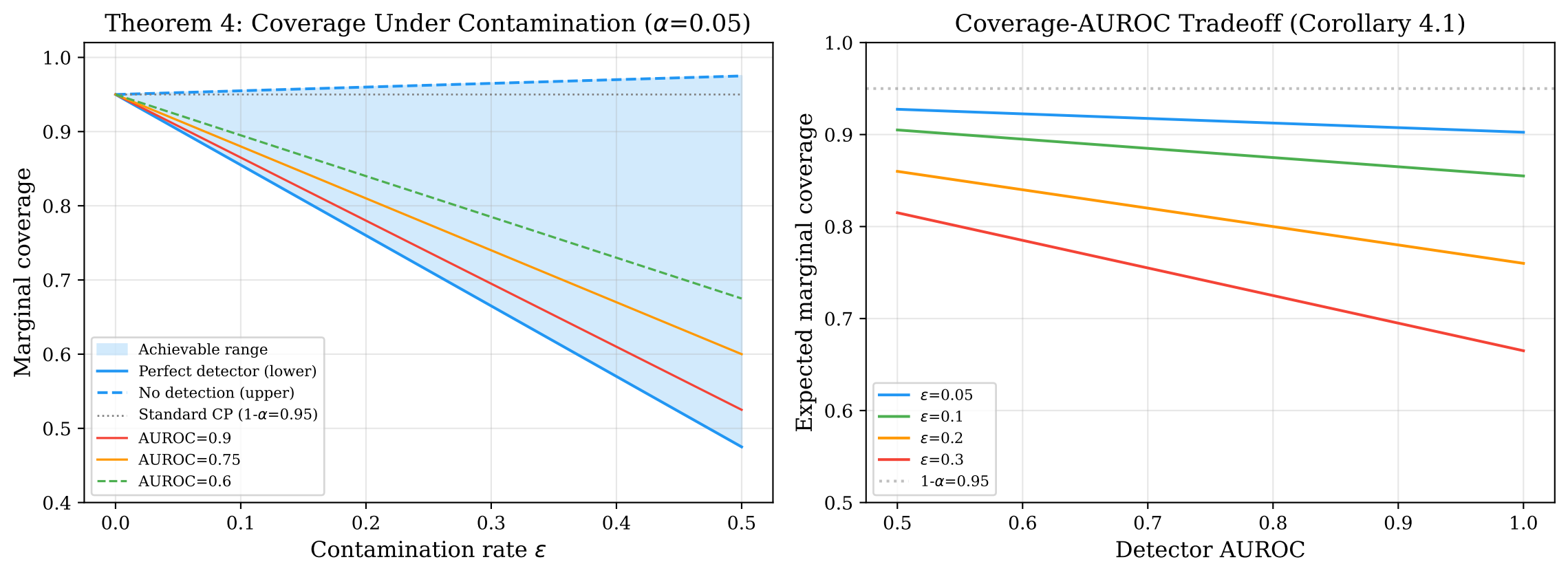
Theorem 4: Coverage Under Contamination (α=0.05)
Marginal coverage
Contamination rate ε
Achievable range
Perfect detector (lower)
No detection (upper)
Standard CP (1-α=0.95)
AUROC=0.9
AUROC=0.75
AUROC=0.6
Coverage-AUROC Tradeoff (Corollary 4.1)
Expected marginal coverage
Detector AUROC
ε=0.05
ε=0.1
ε=0.2
ε=0.3
1-α=0.95